

A PROJECT TITLE

ON

<Hospital Readmission Prediction Using Elastic Net Regularized Logistic Regression >

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

Name P.Pavan Sai roll 2520030417

Name U.Ankith roll 2520030403

Under the Esteemed guidance of

<Dr. Shaik Asif>

<MACHINE LEARNING>

<CSE DEPARTMENT>



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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1. Abstract:

Unplanned hospital readmissions represent a critical challenge for healthcare systems, negatively impacting patient outcomes and driving up operational costs. Accurately identifying patients at high risk of readmission enables healthcare providers to implement proactive, targeted interventions. This project develops a robust predictive model using **Elastic Net Regularized Logistic Regression** to forecast hospital readmissions based on patient clinical profiles and historical health records.

The core methodology implements a dual-penalty framework combining L1 (Lasso) and L2 (Ridge) regularization. This approach effectively manages high-dimensional healthcare datasets by performing automated feature selection to eliminate noise and resolving multicollinearity among correlated clinical variables. The model's performance is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). By balancing feature sparsity with coefficient shrinkage, the model prevents overfitting, resulting in a high-performing and highly interpretable clinical decision-support tool.

2. INTRODUCTION:

2.1 Problem Statement

Predicting hospital readmissions is challenging due to the complex, multi-dimensional nature of electronic health records (EHRs). Clinical datasets often contain a large number of correlated features (multicollinearity) and noisy variables that can cause standard machine learning models to overfit. Therefore, there is a need for a robust predictive framework that can handle high-dimensional data while maintaining high interpretability so that clinicians can understand the primary drivers behind a patient's risk score.

2.2 Proposed Solution

This project proposes the use of **Elastic Net Regularized Logistic Regression**. Standard logistic regression can struggle when predictors are highly correlated or when the feature space is large. Elastic Net overcomes these limitations by combining both L1 (Lasso) penalty—which performs automatic feature selection by shrinking coefficients of irrelevant variables to zero—and L2 (Ridge) penalty—which handles multicollinearity by shrinking coefficients of correlated variables together. This combination provides a stable, accurate, and interpretable solution ideally suited for clinical prediction tasks.

3. SOFTWARE REQUIREMENTS:

3.1 Operating System

- **OS:** Windows 10/11, macOS, or Linux (Ubuntu 20.04 or later recommended)

3.2 Programming Language

- **Python:** Version 3.8 or higher

3.3 Core Libraries & Dependencies

- **Data Manipulation & Numerical Computing:**
 - Pandas (for data loading, cleaning, and preprocessing)
 - NumPy (for numerical array operations)



Signature of the Guide



Course Instructor